DAVIDE BALDELLI

PhD researcher at Polytechnique Montréal and Mila

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EDUCATION

MILA and Polytechnique Montréal *Jan 2025 - Current*
PhD in Computer Engineering (GPA: 4.0)
- Supervisors: Sarath Chandar, Amal Zouaq.
- Courses included Representation Learning, Reinforcement Learning.

University of Bologna *Sept 2021 - Oct 2023*
MSc in Artificial Intelligence (Grade: 110/110 cum laude)
- Courses included Programming for Data Science, Knowledge Representation and Reasoning, Statistical Theory and Methods, Machine Learning, Deep Learning, Computer Vision, Natural Language Processing.
- Thesis: A TWO-step LLM Augmented distillation method for passage Reranking.

University of Florence *Sept 2017 - Mar 2021*
BSc in Mathematics (Grade: 110/110)
- Courses included Analysis, Linear Algebra, Geometry, Probability and Statistics, Logic and Computability.
- Thesis: Big Deviations for Markov Chains.

PUBLICATIONS

Cross-Model In-Context Neurofeedback Reveals (Limited) Activation Control in LLMs *PrePrint*
Davide Baldelli, Robyn Kozierok, Mirko Bronzi, Damiano Fornasiere

ϕTD: Distributional Reinforcement Learning using Characteristic Functions *NeurIPS 2026*
Thomas Mousseau, Tyler Kastner, **Davide Baldelli**, Amir-massoud Farahmand

Probabilistic Calibration Is a Trainable Capability in Language Models *PrePrint*
Davide Baldelli, Sruthi Kuriakose, Maryam Hashemzadeh, Amal Zouaq, Sarath Chandar

LLMs Can't Play Hangman: On the Necessity of a Private Working Memory for Language Agents *CoLLAs 2026 (Oral)* *MemAgents Workshop @ ICLR 2026*
Davide Baldelli, Ali Parviz, Amal Zouaq, Sarath Chandar

CADmium: Fine-Tuning Code Language Models for Text-Driven Sequential CAD Design *TMLR*
Prashant Govindarajan, **Davide Baldelli**, Jay Pathak, Quentin Fournier, Sarath Chandar

TWOLAR: A TWO-Step LLM-Augmented Distillation Method for Passage Reranking *ECIR 2024 (Oral)*
Davide Baldelli, Junfeng Jiang, Akiko Aizawa, Paolo Torroni

POSITIONS AND PROGRAMS

MATS Fellowship (10.0 + 10.1 Extension) *Jun 2026 - Present*
- Supervised by Mirko Bronzi and Damiano Fornasiere from LawZero.
- Investigating introspection capabilities in language models.
- Selected for the MATS 10.1 extension phase (from Sep 2026) to continue the project.

ARENA 7.0 (Alignment Research Engineer Accelerator) *Jan 2026*
- Technical AI safety program focused on building ML engineering skills for alignment research.

Loop AI Labs Inc. *Sept 2022 - Dec 2024*
- Machine Learning Engineer.
- Co-supervised a master's thesis titled "Self Assessment Tool for Medical Exam Candidates" in collaboration with the University of Padova.

National Institute of Informatics (NII) *Mar 2023 - Aug 2023*
- International Internship Program.
- Supervisors: Akiko Aizawa, Paolo Torroni.
- Developing an LLM based text-reranker.

SERVICE AND LEADERSHIP

INF8245AE Machine Learning – Polytechnique Montréal *Fall 2026*
Teaching Assistant

Fatima Al-Fihri Predoctoral Fellowship *Fall 2026*
Research Mentor
- Mentoring a fellow on a machine learning research project through the Fatima Institute for Global AI Research, a program supporting prospective PhD applicants from low- and middle-income countries.

SPAR (Supervised Program for Alignment Research) *Fall 2026*
Research Mentor
- Mentoring two research fellows on building robust activation monitors for language models.

CoLLAs 2026 (Conference on Lifelong Learning Agents) *Nov 2025 - Current*
Web Communication Chair
- Managing the official conference website and digital presence for the 2026 edition.

MILA *Summer 2025*
Teaching Assistant - Summer High School Internships
- Designed and delivered educational workshops on Machine Learning and Probability for high school students.

UILO – Career Orientation Platform *Sept 2024 – Present*
Co-Creator and Technical Lead
- Built UILO, a career orientation platform for post-secondary education with university and career databases and psychometry-based recommendations.

Belle Parole Arci APS *Nov 2023 - Current*
Co-Founder, Non-Profit Association
- Organized community events and communications, built the organization's website **belleparoleaps.it**, and secured grant funding for organizational activities.

ADDITIONAL SKILLS, SERVICE, AND ACHIEVEMENTS

Reviewing	CoLLAs 2026; MemAgent Workshop at ICLR 2026.
Skills	PyTorch, Hugging Face, vLLM, LangChain/LangGraph, Python, Linux, LaTeX, Weights & Biases, Figma.
Recognition	Selected Participant, Y Combinator AI Startup School (Jun 2025).
Languages	English (professional proficiency); Italian (native proficiency).